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DIVISION OF MINERALS & GEOLOGY
RECLAMATION MINING•SAFETY

Date: August 2, 1999

To: Christina Kamnikar

From: Harry Posey

Subject: Review: Report on Storage of Cement Kiln Dust, Southdown, Inc. – Lyons Plant, M-77-208

The captioned operation, a limestone quarry, is permitted to discard cement kiln dust (CKD) from the adjacent Lyons cement plant into the quarry pit. Because high alkalinity and leachable metals occur in some kiln dusts the Division requested that the Operator examine the pH characteristics and metals leachability of the CKD. Based on those results, for parameters that exceeded water quality standards, the operator was to examine the potential effects on groundwater receptors via geochemical and/or hydrological modeling. The operator chose a hydrological modeling approach consisting of mixing/dilution and dispersion.

TESTING AND MODELING

The report is a well-presented rendition of a sufficient geochemical assessment – leachate analyses using an aggressive leach procedure – and a rational to conservative hydrological modeling exercise. The leaching and modeling results are based on conservative assumptions. Results were compared with groundwater standards for the more conservative of either the drinking water or agriculture use.

Solutes from the CKD leach test exceeded water quality standards for several parameters as follows:

Parameter	Standard	Result
Selenium (mg/L)	0.02	0.061
Thallium (mg/L)	0.002	0.014
TDS (mg/L)	400	2050
pH (std units)	6.5-8.5	12.38
Gross alpha (pCi/L)	15	26 ±28
Gross beta (mrem/yr)	4	343±37

The above values served as inputs for mixing and dispersion calculations to simulate potential consequences on the alluvial aquifer and on the groundwater (Dakota Formation) aquifer. The calculations assumed no adsorption or attenuation of metals and no chemical reactions.

Model calculations used a mixture of measured values, rational assumptions, and worst case assumptions. The authors presumed that this approach is conservative. Measured values were input for CKD volume, leach test results and hydraulic conductivity. Rational assumptions were input for hydraulic gradient (actually, a range of values from reasonable to worst case were tested), dispersion coefficient, and saturated thickness of alluvium and Dakota Sandstone. Conservative and/or worst case assumptions were input for CKD cover composition, leachate flux rate, porosity of the CKD, and metal adsorption and attenuation. The model assumed porous media flow rather than fracture flow.

DISCUSSION

On the presumption that the input values for hydraulic gradient and hydraulic conductivity are either rational or conservative, the modeling results do represent a conservative approach. The assumption of porous media flow rather than fracture flow seems reasonable; casual field observations from my earlier site inspection show that the formations are gently folded, have no visible faults, and an unremarkable set of orthogonal fractures with apparently wide spacing. Moreover, the pit floors and walls form a semi-confining zone whereby, according to the operator, evaporation, not flow to groundwater, is the main source of water loss from the pit.

In an actual water rock system such as this, adsorption and/or ion exchange involving clays should remove some metals from solution. The model assumed no adsorption or ion exchange; metals attenuation was achieved solely by dilution. Also, high pH solutions of lime should be buffered by CO_2 and/or clay reactions, yet the model presumed no reactions, only straight line mixing. Overall, these form the basis of a geochemically conservative model because metals and chemicals that would react to attenuate metals and buffer pH in real solutions were presumed to be inert and obey straight line mixing behavior.

Verification of the model results if provided in part through an analysis of the pit water. The pit collects precipitation and dust control water runoff directly from the CKD along with some seepage from the upland irrigation ditch. Except for sulfate, which is higher in the pit water than in the CKD leachate, and thallium and selenium, which are lower in the pit water than in the leachate, the two waters are similar in composition.

The CKD was leached using the Synthetic Precipitation Leach Procedure (SPLP) (EPA Method 1312). This is a physically aggressive test, even for already powdered materials, because the samples are rolled in a vat for 18 hours to assure maximum contact between solids and solutions. The procedure is only moderately chemically aggressive. The high pH from the leachate tests is apparently due to dissolution of unreacted lime in the solids. Lime, stored in the atmosphere, fairly quickly reacts with atmospheric carbon dioxide to form calcium carbonate (limestone) provided air can reach the lime. Because pH from the A-pit sample was similar to groundwater (about pH 8), it appears that where lime is in contact with rainwater, lime is being rapidly neutralized.

CONCLUSION

Despite leach test results which showed water quality exceedances for several parameters, conservative modeling results indicate that water quality effects will be minimal and should not exceed standards. It is recommended that the operation continue with their CKD disposal plan, as permitted.

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